YULING SHEN

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EDUCATION

Dec. 2021	**PhD** in Aerospace Engineering, Texas A&M University
Dec. 2016	**M.S.** in Mechanical Engineering, Northwestern University
Jun. 2015	**B.S.** in Mechanical Engineering, University of California San Diego

ACADEMIC APPOINTMENT

2023–now	Associate Professor, School of Future Science and Engineering, Soochow University, China

RESEARCH FUNDING AND PROJECTS

2026–2028	(PI) National Natural Science Foundation of China (NSFC), Young Scientists Fund — CNY 300,000.
2024–2027	(PI) Natural Science Foundation of Jiangsu Province, Young Scientists Fund — CNY 200,000.
2025–2027	(PI) Jiangsu University–Industry Collaborative Research Project — CNY 300,000.
2023–2029	(PI) Excellent Young Scientist Start-up Research Fund, Soochow University — CNY 400,000.
2019–2020	(Doctoral researcher) NASA Innovative Advanced Concepts (NIAC), *Lunar-Polar Propellant Mining Outpost (LPMO): Affordable Exploration and Industrialization*.
2017–2018	(Doctoral researcher) NASA Innovative Advanced Concepts (NIAC), *Tensegrity Approaches to In-Space Construction of a 1g Growable Habitat*.

ACADEMIC SERVICE

2025–2027	Young Editorial Board Member, *Advances in Astronautics*.
2025–2030	Council Member, Jiangsu Society of Aeronautics and Astronautics.

PUBLICATIONS

* Corresponding author

Dynamics Modeling

1. Chen, M., **Shen, Yuling** & Skelton, R. E. Model-Based and Markov Data-Based Linearized Tensegrity Dynamics and Analysis of Morphing Airfoils. In *AIAA SCITECH 2024 Forum* AIAA Paper 2024-2347 (2024).
2. **Shen, Yuling**, Chen, M.* & Skelton, R. E. A Markov data-based approach to system identification and output error covariance analysis for tensegrity structures. *Nonlinear Dynamics* **112,** 7215–7231 (2024).
3. **Shen, Yuling***, Chen, M., Majji, M. & Skelton, R. E. An Efficient q-Markov Covariance Equivalent Realization Approach to System Identification. In *IFAC-PapersOnLine* 22nd IFAC World Congress. **56,** 108–113 (2023).
4. **Shen, Yuling***, Chen, M., Majji, M. & Skelton, R. E. Q-Markov Covariance equivalent realizations for unstable and marginally stable systems. *Mechanical Systems and Signal Processing* **196,** 110343 (2023).

Dynamics Simulation

5. Shi, L., Cao, W., Chen, M. & **Shen, Yuling**[*]. Markov Parameters Generation for Data-based Modeling of Tensegrity Robots Considering Finite Word-Length Effects. In *2025 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)* 3831–3837 (IEEE, 2025).
6. **Shen, Yuling**, Chen, M.[*] & Skelton, R. E. Finite word-length optimal simulation for high-dimensional dynamical systems: Examples of tensegrity structures. *Mechanical Systems and Signal Processing* **215,** 111415 (2024).

Dynamics Control

7. Chen, M.[*] & **Shen, Yuling**[*]. Resource-efficient covariance control with optimal sensor and actuator selection for tensegrity structures. *Engineering Structures* **362,** 122908 (2026).
8. **Shen, Yuling**[*], Chen, M., Habtour, E. & Skelton, R. E. Sensor Fault Detection Approach to Tensegrity Structures Using Markov Parameters. In *Earth and Space 2024* 584–593 (ASCE, 2024).
9. **Shen, Yuling**, Chen, M.[*] & Skelton, R. E. Markov data-based reference tracking control to tensegrity morphing airfoils. *Engineering Structures* **291,** 116430 (2023).

Structural Design

10. Su, L., Chen, M. & **Shen, Yuling**[*]. Inverse-Kinematic Design of Deployable Scissor Mechanisms along Arbitrary Curves. In *77th International Astronautical Congress (IAC 2026)* Antalya, Türkiye. Accepted for oral presentation; Paper ID IAC-26,C2,2,9,x111788 (2026).
11. Su, L., Chen, M. & **Shen, Yuling**[*]. Parametric Studies of Hoberman Topology for the Design of Scalable and Deployable Structures. In *76th International Astronautical Congress (IAC 2025)* IAF Materials and Structures Symposium, Sydney, Australia, 143–152 (International Astronautical Federation, 2025).
12. Zhang, G., Chen, M.[*], Chen, D. & **Shen, Yuling**[*]. Lightweight designs of simply supported tensegrity structures and their applications to bridges. *Composite Structures* **357,** 118923 (2025).
13. Chen, M., **Shen, Yuling**, Goyal, R., Majji, M. & Skelton, R. E. Energy analysis of growth adaptable artificial gravity space habitat. In *2018 AIAA SPACE and Astronautics Forum and Exposition* AIAA Paper 2018-5109 (2018).

PATENTS

P1. **Shen, Yuling**, Wang, W., Chen, T., Fan, C., Li, X. & Zhu, M. *A spatial swing-angle measurement system, method and device for suspended low-gravity simulation*. Chinese invention patent ZL2026100679354, granted 2026.

P2. **Shen, Yuling**, Su, L., Wang, W. & Zhang, G. *A Hoberman ring and its design method and applications*. Chinese invention patent ZL2025120523857, granted 2026.

P3. **Shen, Yuling**, Wang, W., Chen, T., Fan, C., Li, X. & Zhu, M. *A control method, system and electronic device for suspended low-gravity simulation*. Chinese invention patent ZL2026100679797, granted 2026.

P4. Quan, Y., **Shen, Yuling**, Zhao, Y. & Cheng, Y. *A data output method and system based on timbre and emotion simulation*. Chinese invention patent ZL2024117394697, granted 2025.

P5. **Shen, Yuling**, Wang, W. & Zhang, G. *An energy-efficient pressure-control method and system for cleanrooms using dynamic pipe networks and model predictive control*. Chinese invention patent ZL202510534007X, granted 2025.

RESEARCH SOFTWARE

TsgDMC: an open-source toolbox for data-based modelling and control of deployable tensegrity structures.

PRESENTATIONS

1. 76th International Astronautical Congress (IAC), Sydney, Australia, 2025.

2. IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), Hangzhou, China, 2025.
3. Chinese Congress of Mechanics, Changsha, China, 2025.
4. 17th National Conference on Nonlinear Dynamics and Motion Stability, Nanning, China, 2025.
5. Society of Engineering Science Technical Meeting (SES), Hangzhou, China, 2024.
6. 7th Cross-Strait Symposium on Dynamics, Vibration and Control, Hong Kong and Macao, China, 2024.
7. AIAA SciTech Forum, Orlando, FL, USA, 2024.
8. ASCE Earth and Space, Miami, FL, USA, 2024.
9. 22nd IFAC World Congress, Yokohama, Japan, 2023.
10. European Control Conference (ECC), Bucharest, Romania, 2023.
11. (Poster) *Tensegrity System Research Snapshots at Land, Air, and Space Robotics (LASR) Laboratory*, TAMU Physics & Engineering Festival, 2021.
12. (Poster) *In-Space Construction of a 1g Growable Habitat*, TAMU Physics & Engineering Festival, 2019.
13. AIAA SPACE and Astronautics Forum and Exposition, Orlando, FL, USA, 2018.
14. (Poster) *Fluidic Treadmill System*, UCSD MAE Senior Design Poster Session, 2015.

MEDIA

1. Phys.org, 2020.
2. Voice of America, 2019.
3. Texas A&M Engineering News and Video, 2019.
4. The Wall Street Journal, 2018.

AWARDS AND HONOURS

1. Nomination Award for Young Scientists, awarded by Jiangsu Society of Aeronautics and Astronautics, 2025.
2. First Prize of Teaching Innovation Competition, awarded by Soochow University, 2025.
3. 2022–2023 Suzhou Outstanding Academic Paper in Natural Sciences, 2024.
4. AERO Graduate Excellence Fellowship, awarded by Texas A&M University, 2021.
5. American Society of Mechanical Engineers (ASME) Award, awarded by University of California San Diego, 2015.

TEACHING AND SUPERVISION

Course instructor, Soochow University

1. Complex Analysis (Fall 2023)
2. Numerical Analysis (Spring 2024, Spring 2025, Fall 2025, Spring 2026)
3. Modern Control Theory (Fall 2024, Fall 2025)
4. Professional English (Fall 2024, Fall 2025)
5. Data-Driven Dynamic Modelling (Spring 2025)

Supervision

1. Supervised 11 master's students in Mechanical Engineering, Computer Science, and Artificial Intelligence.
2. Supervised nine undergraduate senior design projects in Mechanical Engineering and Artificial Intelligence.
3. Advised 18 undergraduate professional internships.